EST200 DESIGN & ENGINEERING

Module – 1

1. ***Discuss the importance of design constraints?***
   Design constraints are like rules that help designers make good choices. They give direction and help designers focus on what's important. Constraints can also inspire new ideas by pushing designers to be creative within limits. They help manage resources like time and money, ensuring the final product is realistic. Constraints keep designs from being too complicated and help ensure quality. They also encourage teamwork among different groups involved in the design process. In a nutshell, design constraints are important because they guide designers, spark creativity, manage resources, maintain quality, and promote collaboration.

2. ***Describe how to select the "best possible design" from the generated design alternatives.***

- **Identify Criteria**: First, figure out what's most important for your project. Is it cost, functionality, aesthetics, durability, or something else?
- **Rank Criteria**: Once you know what matters most, rank these criteria in order of importance. For example, if you're designing a car, safety might be more important than color options.
- **Evaluate Designs**: Look at each design option and see how well it meets each criterion. Give each design a score based on how well it meets your needs.
- **Compare Scores**: Add up the scores for each design. The one with the highest total score is likely your best bet. But also consider if any design has major flaws or if there's a tie, and adjust accordingly.
- **Final Check**: Before making your decision, take a step back and make sure the chosen design aligns with your overall goals and requirements.
- **Feedback Loop**: Sometimes, it's helpful to get input from others or to revisit your criteria and scores to make sure you're making the right choice. Don't be afraid to iterate if needed.

The "best" design isn't always perfect, but it's the one that best meets your needs and goals within the given constraints.

3. ***Outline the significance of understanding customer requirements in design process***
   - **Customer Satisfaction**: If you don't know what your customers want, you might end up making something they don't like or need. Understanding their requirements helps ensure they'll be happy with the final product.
   - **Quality Assurance:** Knowing what your customers need helps you create a product that meets their expectations. This reduces the chances of errors or rework later on.
   - **Cost Efficiency**: Designing based on customer requirements means you're not wasting time and money on features or details that aren't necessary. This can save resources and make the design process more efficient.
   - **Competitive Edge**: By understanding what customers want and need, you can create products that stand out from the competition. This gives you an edge in the market and can help attract more customers.

- **Building Trust:** When customers see that you're listening to their needs and preferences, they're more likely to trust your brand and become repeat customers. This can lead to long-term success and loyalty.

Overall, understanding customer requirements is like having a map that guides you to create products that people actually want and love. It's the foundation of a successful design process.

4. ***Describe any three constraints that can occur in design process of a lunch box***

Size and Shape Constraints: The lunch box needs to be a certain size and shape to accommodate the amount of food it's intended to hold and fit into standard lunch bags or backpacks.
Eg: Size of lunch box should be oval, there should be separation for placing curry

Material Constraints: The choice of materials for the lunch box can be constrained by factors such as durability, safety, and cost.
Eg: It should be made of steel, It should made of some materials which can withstand high temperature.

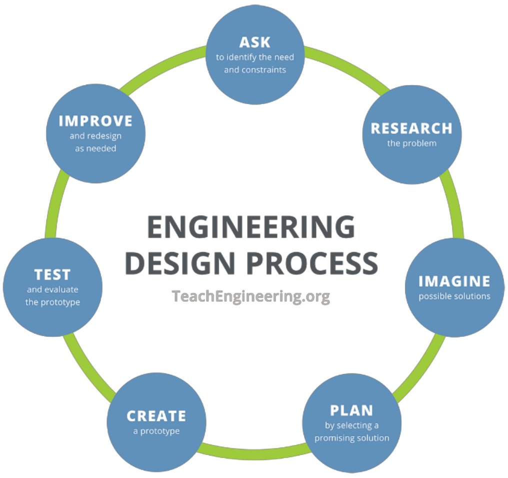
Insulation and Thermal Constraints: If the lunch box is intended to keep food hot or cold for extended periods, there are constraints related to insulation and thermal performance.
Eg: It should keep food hot for 5 hours

5. How to identify the customer requirements of design?
   - **Communicate with the Customer:** Initiate discussions with the customer to gather their input. This can be done through meetings, interviews, surveys, or feedback forms. Actively listen to their ideas, concerns, and expectations.
   - **Ask Open-Ended Questions**: Instead of leading questions, ask open-ended questions that encourage detailed responses. For example, "What features are most important to you?" or "How do you envision using this product/service in your daily life?"
   - **Analyze Existing Data:** Review any existing data, such as market research reports, customer feedback, or previous design iterations. This can provide valuable insights into customer preferences and pain points.
   - **Create Personas:** Develop personas representing different segments of your target audience. These personas should encapsulate typical customer characteristics, behaviors, and goals. Use these personas as a reference point when considering design choices.
   - **Conduct User Testing:** Prototype your design ideas and conduct user testing sessions with representative customers. Observe how they interact with the prototype and gather feedback on what works well and what needs improvement.
   - **Use Design Thinking Methods**: Employ design thinking methodologies, such as empathy mapping or journey mapping, to gain a deeper understanding of the customer experience. This can uncover latent needs and opportunities for innovation.

- **Prioritize Requirements:** Not all customer requirements are equally important. Work with the customer to prioritize their needs based on factors like feasibility, impact, and cost.

- **Iterate and Refine:** Design is an iterative process. Continuously gather feedback from customers throughout the design process and incorporate it into subsequent iterations. This ensures that the final design meets their requirements effectively.

6. ***What is engineering design? Draw a diagram to represent the engineering design process?***
   Engineering design is the systematic process of creating innovative solutions to problems or fulfilling needs through the application of scientific and mathematical principles.



7. ***Explain and differentiate conceptual design and detailed design?***

Conceptual Design:

1. **Purpose**: Conceptual design is the initial phase of the design process, where the primary goal is to explore and generate broad ideas and concepts to address the problem or need identified. It's about brainstorming and creating a range of potential solutions without getting into the specifics of how they will be implemented.
2. **Level of Detail**: At this stage, the level of detail is relatively low. The focus is on generating ideas, defining high-level requirements, and exploring different approaches to solving the problem. Sketches, diagrams, and rough prototypes may be used to convey concepts.
3. **Flexibility**: Conceptual design allows for maximum creativity and flexibility. Engineers and designers are encouraged to think outside the box and explore unconventional solutions without being constrained by practical considerations.
4. **Evaluation**: Concepts are evaluated based on their feasibility, innovation, and alignment with project objectives and requirements. This evaluation helps narrow down the options and select the most promising concepts for further development.

Detailed Design:

1. **Purpose**: Detailed design occurs after a concept has been selected and serves to flesh out the chosen solution in greater detail. The focus shifts from generating ideas to refining and specifying how the chosen concept will be realized.
2. **Level of Detail**: Detailed design involves specifying the dimensions, materials, components, and manufacturing processes required to bring the concept to life. Engineers create detailed drawings, models, and specifications that provide clear instructions for implementation.
3. **Precision**: Detailed design requires a high level of precision and accuracy. Every aspect of the design must be carefully considered and specified to ensure that the final product meets the desired performance, quality, and safety standards.
4. **Implementation Guidance**: Detailed design provides the necessary information and guidance for the actual implementation of the solution. It serves as a blueprint that manufacturers, contractors, or builders can follow to produce the final product.

Differentiation:

1. **Focus**: Conceptual design focuses on generating ideas and exploring possibilities, while detailed design focuses on refining and specifying a chosen concept.
2. **Level of Detail**: Conceptual design is characterized by low detail and flexibility, while detailed design involves high detail and precision.
3. **Purpose**: Conceptual design helps identify potential solutions and select the most promising concepts, while detailed design provides the information needed to implement the chosen solution effectively.

8. ***State how engineering design is different from other kinds of design.***

   Engineering design is different from other types of design because it's all about solving practical problems using science and math.

   For example, while graphic designers might focus on making things look nice, engineers focus on making things work properly.

   Engineers need to think about lots of technical stuff, like how things will be built, what materials to use, and whether the design will be strong enough to handle its job.

   Unlike some other types of design, engineering design also has to consider things like safety rules, regulations, and making sure the design will last a long time without breaking.

   So, while design in general is about creating something, engineering design is about creating something that works reliably and meets specific needs in the real world.

9. ***Define engineering design.***

   Engineering design is like solving a puzzle but with real-world stuff. It's about using science and math to come up with smart solutions to problems, like making machines, buildings, or systems that work well and do what they're supposed to do. So, engineering design is basically finding clever ways to build things that help people in their everyday lives.

10. ***What is the importance of systematic design.***

    Systematic design is like following a recipe when you're cooking. It helps keep everything organized and makes sure you don't forget any important steps. In engineering, systematic design means having a clear plan and following a step-by-step process to create something new, like a building or a machine.

The importance of systematic design is that it helps prevent mistakes and ensures that the final product turns out the way it's supposed to. Just like following a recipe helps you make a delicious meal, following a systematic design process helps engineers create things that work well and meet all the requirements. It also makes it easier to fix problems if they come up along the way. So, systematic design is like having a roadmap to success in engineering projects.

11. How is engineering design different from other kinds of design?

Purpose: Engineering design focuses on solving practical problems or needs using science and math, while other kinds of design may focus more on aesthetics or user experience.
Technical Complexity: Engineering design deals with complex systems and requires deep knowledge of physics, mechanics, and other technical fields, whereas other design fields may not require as much technical expertise.
Functionality vs. Appearance: Engineers prioritize making things work reliably and efficiently, while other designers might focus more on how things look or feel.
Constraints: Engineers have to consider technical limitations, regulations, and budget constraints when designing, while other designers may have different sets of constraints, such as artistic vision or market trends.
In simple terms, engineering design is all about making things work effectively and safely, while other kinds of design might prioritize appearance or user experience.

12. Enumerate the two objectives and three constraints in the design of a table fan

Objectives:

1. **Functionality**: The primary objective of a table fan design is to effectively circulate air to provide cooling in a room or space.
2. **User Convenience**: Another objective is to ensure user convenience and ease of use, including features like adjustable speed settings and oscillation.

Constraints:

1. **Safety**: Safety is a critical constraint in table fan design. It includes ensuring that the fan blades are properly enclosed to prevent accidents, and electrical components meet safety standards to avoid electrical hazards.
2. **Cost**: Cost is an important constraint in the design process. Designers must balance the inclusion of features and materials to ensure the final product remains affordable and competitive in the market.
3. **Size and Portability**: Table fans are designed to be compact and portable, so constraints related to size and weight must be considered. The fan should be small enough to fit on a table or desk while still providing sufficient airflow, and it should be lightweight enough for easy transportation or relocation

13. List out three methods to select the best concept from many at the initial stage of Design

- **Voting or Survey**: Each team member or stakeholder can vote or provide feedback on the concepts they think are the strongest. The concept with the most votes or highest overall rating is selected. It's like when you and your friends vote on which movie to watch – the one with the most votes wins!

- **Decision Matrix**: Create a decision matrix where you list all the concepts along the top and criteria (like cost, effectiveness, ease of implementation) along the side. Then, rate each concept against each criterion. Add up the scores for each concept to see which one scores the highest overall. It's similar to making a pros and cons list for different options.
- **Prototype and Testing**: Build quick and simple prototypes of the different concepts. Then, test them out to see which one works best in practice. For example, if you're designing a new type of cup holder for a car, you might make prototypes of a few different designs and see which one holds the cup most securely without spilling. It's like trying out different recipes to see which one tastes the best.

PART – B Questions

14. ***You are assigned to design an automatic sanitizer dispenser for your institution. Set the objective and constraints. Apply the standard design procedure to arrive the final concept.***
15. ***Explain the method of identifying the customer requirement of an umbrella and the method of finalising the product specification.***
16. ***Design a study table for a child upto 5 years old specifying objectives, functions and constraints through various stages of design processes. Use hand sketches to illustrate the process.***
17. ***Choose the best design for a laptop stand for avoiding overheating, also incorporate the customer requirements. Show how design objectives were finalized considering the constraints also.***
18. ***Explain in detail the different stages of a design process.***
19. ***Find the customer requirements for designing a new wrist watch. Identify the design constraints. Explain, how were the design objectives finalized considering the design constraints? Use hand sketches.***
20. ***Find the customer requirements for designing a website for an educational institution. Show how the design objectives were finalized considering the design constraints. Sketch a layout of the website showing dropdown menus.***
21. ***Show the designing of an iron box going through the various stages of the design process. Use hand sketches to illustrate the processes.***
22. ***Explain the design process through designing a handbag for women of age group of 15 to 25 years. Use hand sketches to support your idea***
23. ***Describe the concept of generating design alternatives and choosing a design through designing a coffee mug with the help of sketches***
24. ***Design two alternatives of a chair suitable for a five-year-old child, and then to narrow down to the best design based on objectives and constraints. Sketch both the designs.***
25. ***Identify the objectives, functions and constraints for designing a water level indicator. Illustrate the various stages of the design process. Provide suitable sketches.***

Module – 2

1. ***How does the design thinking approach help engineers in creating innovative and efficient designs?***

Design thinking helps engineers create innovative and efficient designs by focusing on understanding the needs of the people who will use the product or system. Here's how it works in simple terms:

1. **Empathize**: Engineers start by putting themselves in the shoes of the people who will use their design. They talk to users, observe how they interact with similar products, and try to understand their needs, preferences, and challenges. It's like trying to see the world through someone else's eyes.
2. **Define**: Once they understand the users' needs, engineers define the problem they're trying to solve. They identify the key goals and constraints that will shape their design. It's like figuring out the rules of a game before you start playing.
3. **Ideate**: Engineers brainstorm lots of different ideas for how they could solve the problem. They encourage creativity and think outside the box, coming up with as many ideas as possible, even if some of them seem a little crazy. It's like brainstorming with friends to come up with ideas for a fun weekend activity.
4. **Prototype**: Engineers build quick and simple prototypes of their ideas to see how they work in practice. These prototypes don't have to be perfect – they're just a way to test out different ideas and see what works and what doesn't. It's like building a rough sketch of a drawing before you add all the details.
5. **Test**: Engineers test their prototypes with real users to get feedback. They see how people interact with the design and use their feedback to make improvements. It's like trying out a new recipe on your friends and asking them what they think so you can make it even better next time.

By following the design thinking approach, engineers can create designs that are not only technically sound but also meet the needs of the people who will use them, leading to more innovative and efficient solutions.

2. ***Explain convergent questioning in design thinking***

Convergent questioning in design thinking is like using a magnifying glass to focus on finding the best solution to a problem. For example:

Imagine you have lots of ideas for solving a problem, like how to make a better backpack for school. Convergent questioning helps you narrow down all those ideas and focus on the ones that are the most promising.

So, instead of asking questions that generate even more ideas, convergent questioning helps you ask questions that help you choose the best ideas. It's like asking:

- Which idea is the simplest to make?
- Which idea solves the problem the most effectively?
- Which idea is the easiest for kids to use?

These questions help you zoom in on the best ideas and decide which one to move forward with. It's like finding the perfect piece of a puzzle that fits just right!

3. ***Explain how conflict in team environment helps in better design of products***

   Different Perspectives: Imagine you and your friends are trying to decide what game to play. One friend wants to play soccer, another wants to play basketball, and you want to play tag. Each person has a different idea of what's best. Similarly, in a design team, different team members might have different ideas about how to design a product. Conflict happens when these ideas clash.

   Debate and Discussion: Conflict leads to debate and discussion. You and your friends might talk about the pros and cons of each game before deciding. Similarly, in a design team, people debate and discuss different design ideas. They talk about what they like and don't like about each idea, and why they think one idea might be better than another.

   Creativity and Innovation: Through this debate and discussion, new ideas can emerge. Maybe you decide to combine elements of soccer and tag to create a new game. Similarly, in a design team, conflicting ideas can lead to new and innovative design solutions that nobody had thought of before.

   Better Solutions: By considering different perspectives and exploring conflicting ideas, teams can ultimately come up with better solutions. Just like how your combined game might be more fun and exciting than any of the individual games, the final product design might be more innovative and effective because it incorporates ideas from different team members.

So, conflict in a team environment can actually be a good thing for designing products because it leads to debate, discussion, and ultimately, better and more creative solutions.

4. ***Describe the iterative process involved in design thinking approach.***

   Empathize: This stage is all about understanding the problem from the perspective of the user or customer. It's like putting yourself in their shoes to see the world from their point of view. You might observe their behaviors, interview them, or even try to experience their challenges firsthand. For example, if you're designing a new backpack for students, you might talk to students about what they like and dislike about their current backpacks.
   Define: Once you understand the user's needs and challenges, you define the problem you're trying to solve. This involves distilling all the information you gathered during the empathy stage into a clear problem statement. It's like writing down the main challenge you need to overcome. Using the backpack example, you might define the problem as: "How might we design a backpack that is comfortable to wear, has enough space for books and supplies, and is durable enough to withstand daily use?"
   Ideate: Now it's time to brainstorm ideas for solving the problem. This stage is all about generating as many creative solutions as possible, without worrying too much about feasibility or practicality. It's like a brainstorming session where you throw out lots of different ideas, no matter how wild they might seem. For the backpack design, you might come up with ideas like adding adjustable straps, incorporating built-in organizers, or using lightweight but durable materials.
   Prototype: Once you have some promising ideas, you create prototypes to bring those ideas to life in a tangible way. Prototypes can be simple and rough, like sketches or cardboard models, or more advanced, like 3D-printed prototypes. The goal is to quickly test your ideas and see how they work in the real world. For the backpack design, you

might create prototypes of different strap designs or material combinations to see which ones are most comfortable and durable.

Test: With your prototypes in hand, you test them with real users to gather feedback. This stage is all about learning from how users interact with your prototypes and using that feedback to refine your designs. It's like conducting experiments to see what works and what doesn't. For the backpack design, you might ask students to wear the prototypes and provide feedback on comfort, usability, and durability.

Iterate: Finally, based on the feedback you receive during testing, you go back to the drawing board and make improvements to your designs. This might involve refining existing ideas, discarding ideas that didn't work, or coming up with entirely new solutions. The process then repeats itself, with each iteration bringing you closer to a final, optimized solution. It's like a continuous cycle of learning and improvement until you arrive at the best possible design for your users.

These stages of design thinking – empathize, define, ideate, prototype, test, and iterate – form a flexible and iterative framework for approaching complex design challenges. By following these stages, designers can create innovative solutions that truly meet the needs of their users.

5. ***Describe the importance of empathize phase in design thinking.***

- **Understanding Needs**: When you empathize with the people who will use your product or service, you get to know them better. It's like asking your friend how they're feeling so you can understand what they need. By talking to people, observing them, and experiencing things from their perspective, you learn about their problems, challenges, and desires.
- **Inspiring Solutions**: Empathy helps you come up with better solutions to their problems. It's like when you see your friend struggling with homework, you might come up with a fun way to study together. By understanding what people really care about and what frustrates them, you can create designs that truly address their needs and make their lives better.
- **Building Trust**: When people feel like you understand them, they're more likely to trust you and your designs. It's like when your friend knows you always listen to them, they trust you to help them with their problems. By showing empathy, you build a connection with your users and show them that you care about making their lives easier and more enjoyable.

6. ***How prototype will help to identify the best possible solution for the problem?***

- **Test Before Commitment**: A prototype is like a rough draft of your idea. It's not perfect, but it gives you a chance to try out different solutions without fully committing to one. Just like how you might try out different plays in practice to see which one works best.
- **See What Works**: When you test your prototype, you get to see how well it works in real life. Maybe your first idea seemed great on paper, but when you try it out, you realize it doesn't work as well as you thought. Testing helps you see what works and what doesn't, just like how trying out different moves in practice helps you see which ones are most effective.
- **Get Feedback**: Testing your prototype also lets you get feedback from others. Maybe your teammates have suggestions for how to improve your idea, or maybe they notice things you didn't think of. Their input helps you refine your solution and make it even better.
- **Iterate and Improve**: Based on what you learn from testing and feedback, you can make changes to your prototype and try again. It's like refining your technique in practice until you have it just right. Each iteration brings you closer to the best possible solution for the problem you're trying to solve.

7. ***Compare convergent thinking and divergent thinking?***

Convergent Thinking:	Divergent Thinking:
Focus: Convergent thinking is about narrowing down options and focusing on finding the single best solution to a problem. Logic and Analysis: It involves using logic and analysis to evaluate different ideas and select the most promising one. It's like using a magnifying glass to zoom in on the best solution. Single Solution: Convergent thinking typically leads to a single solution or answer that best meets the criteria or requirements of the problem. Decision Making: It's useful for making decisions and solving problems where there is a clear goal or right answer	Exploration: Divergent thinking is about exploring as many possible options or ideas as you can, without worrying about whether they're good or bad. Creativity and Imagination: It involves using creativity, imagination, and lateral thinking to generate a wide range of ideas. It's like throwing paint at a canvas and seeing what shapes emerge. Multiple Solutions: Divergent thinking often leads to multiple solutions or possibilities, some of which may be unconventional or unexpected. Brainstorming: It's useful for brainstorming, exploring new possibilities, and generating innovative solutions to complex problems where there may not be a clear answer.

8. ***How to perform design thinking as a team managing the conflicts?***

- **Be Kind and Open**: Design thinking is like working on a project with your friends. Everyone has different ideas, but it's important to be kind and listen to each other's thoughts without getting upset.
- **Have a Goal**: Just like when you play a game, there's a goal you're trying to reach. Make sure everyone knows what problem you're trying to solve and what you want to achieve.
- **Listen to Everyone**: Everyone's ideas are important, even if they're different from yours. Listen carefully to what your friends say and try to understand where they're coming from.
- **Work Together**: Design thinking is like putting together a puzzle – each person's ideas fit together to create the solution. Try to find ways to combine everyone's ideas to make something great.
- **Stay Positive**: Sometimes, there might be disagreements or problems, but it's important to stay positive and keep trying. Remember, it's okay to make mistakes – that's how we learn and grow.
- **Celebrate Success**: When you find a solution that works, celebrate together as a team. It's like winning a game – you did it together, and that's something to be proud of!

9. ***What are the most popular techniques used in convergent thinking.***
   - Decision Matrix
   - Force Field Analysis

- Weighted Scoring Model
- Pareto Analysis
- Criteria Decision Analysis

10. Describe three steps to facilitate design thinking in your team.

- **Create a Collaborative Environment**:
  - Encourage open communication and active participation from all team members.
  - Foster a culture where diverse perspectives are valued and respected.
  - Provide opportunities for brainstorming sessions and idea generation.
- **Define Clear Goals and Objectives**:
  - Clearly define the problem or challenge you're trying to solve.
  - Establish goals and objectives for the design thinking process.
  - Ensure that everyone understands the scope and expectations of the project.
- **Implement Design Thinking Methods and Tools**:
  - Introduce design thinking methods and tools to guide the team through the process.
  - Use techniques such as empathy mapping, journey mapping, and prototyping to gain insights, generate ideas, and test solutions.
  - Provide training and support to team members as needed to familiarize them with design thinking concepts and methodologies.

11. How does design thinking approach help engineers in creating innovative and efficient designs?

Design thinking helps engineers create innovative and efficient designs by emphasizing understanding user needs, generating diverse ideas, prototyping and testing solutions, and iterating based on feedback. By focusing on empathy and user-centered design, engineers can develop products and systems that truly address the needs and preferences of their intended users. The iterative nature of design thinking allows engineers to explore multiple solutions, refine their designs based on real-world testing, and continuously improve until they achieve the best possible outcome. This approach fosters creativity, collaboration, and a deeper understanding of user requirements, ultimately leading to more successful and impactful design solutions.

12. Differentiate conventional thinking and creative thinking

Conventional Thinking:	**Creative Thinking**:
• Conventional thinking relies on established norms, rules, and traditional methods to solve problems. • It follows predictable and linear patterns, often relying on past experiences and established solutions. • Conventional thinking tends to prioritize efficiency and practicality over innovation and novelty.	• Creative thinking involves generating new and original ideas by breaking away from conventional thought patterns. • It encourages exploring unconventional solutions, thinking outside the box, and embracing ambiguity. • Creative thinking values imagination, intuition, and divergent thought processes.

• It may be more structured and systematic, focusing on logical reasoning and analytical processes.	• It may involve experimentation, risk-taking, and openness to failure as part of the creative process.

13. ***List the techniques to improve the thinking process in a design team***
    - Brainstorming
    - Mind Mapping
    - Storyboarding
    - Prototyping
    - Role Playing
    - Collaborative Workshops
    - Design Critiques
    - User Feedback Sessions

PART B Questions

14. ***Design a water bottle that can be opened with one hand. Illustrate the various stages involved in design thinking. Sketch the final design.***
15. ***Illustrate the design thinking process through designing a walking stick for elderly people.***
16. ***Design a parachute mechanism for safe landing an egg which is dropped from a height of 3 meters using iterative design thinking process with the help of sketches***
17. ***Illustrate the design thinking approach for designing a wearable technology for a college student. Describe each stage of the process. Illustrate the solution using sketches.***
18. ***Explain the five different stages of design thinking? Illustrate it with the help of a face mask design***
19. ***Show how divergent and convergent thinking process will help to choose the best design from a list of possible solutions, considering water jug as a case study.***
20. ***Design a suitable product for easy cleaning of dust from the chalkboards in a classroom. Illustrate the various stages involved in design thinking. Sketch the final design.***
21. ***Explain the five stages of design thinking process in detail and justify why design thinking is a nonlinear process.***
22. ***Create three numbers of possible designs for a pocket-sized umbrella and then refine them to the best design using divergent thinking and convergent thinking***
23. ***Design a carry bag for shopping purposes. Illustrate the various stages involved in design thinking. Sketch the final design.***
24. ***Construct a number of possible designs and then refine them to narrow down to the best design for a baby seater in a car. Show how convergent-divergent thinking helps in the process. Illustrate with hand sketches.***
25. ***Illustrate the design thinking approach for designing a study table for kids within a limited budget. Describe the design thinking process.***
26. ***Illustrate three possible conceptual designs of a writing aid for handless person and refine the concepts to obtain the best solution.***

Module – 3

1. ***Explain the factors to be considered in preparing technical reports to communicate a design efficiently?***

When preparing technical reports to communicate a design efficiently, several factors need to be considered to ensure clarity, accuracy, and effectiveness. Here are some key factors:

Audience: Understand who will be reading the report and tailor the content and language to their level of expertise. Consider whether the audience is technical or non-technical and adjust the level of detail accordingly.

Purpose: Clarify the purpose of the report. Understanding the purpose will help you structure the content appropriately and prioritize information.

Clarity: Use clear and concise language to explain technical concepts. Define technical terms when necessary and use illustrations, diagrams, and examples to enhance understanding.

Structure: Organize the report logically with clear headings and subheadings. Use a consistent format throughout the document to guide the reader through the content.

Technical Content: Provide detailed information about the design, including specifications, methodologies, and calculations. Present data and results accurately.

Visual Aids: Use visual aids such as charts, graphs, and diagrams to illustrate key points and enhance comprehension.

Accuracy: Verify the accuracy of all technical information presented in the report. Double-check calculations, measurements, and references to ensure correctness.

Readability: Pay attention to the readability of the report by using appropriate font sizes, styles, and formatting. Break up long paragraphs into smaller chunks, and use bullet points or numbered lists for clarity.

Review and Revision: Before finalizing the report, review it carefully for errors, inconsistencies, and clarity. Consider seeking feedback from colleagues or experts in the field and incorporate any suggestions for improvement.

2. ***Explain the role of mathematics and physics in the design engineering process.***

Mathematics and physics play fundamental roles in the design engineering process, providing the analytical foundation and framework for understanding, modeling, and optimizing systems. Here's how each discipline contributes to the design engineering process:

Mathematics:

Modelling and Analysis: Mathematics provides tools for modelling complex engineering systems and analysing their behaviour. Differential equations, linear algebra, calculus, and numerical methods are commonly used to develop mathematical models that describe the relationships between different variables in a system.

Optimization: Mathematical optimization techniques are applied to maximize performance, minimize costs, or achieve other design objectives. Optimization algorithms, such as linear programming, nonlinear optimization, and evolutionary algorithms, help engineers find the best design solutions within given constraints.

Simulation and Prediction: Mathematics is used to simulate the behaviour of engineering systems through computational modelling. Simulation allows engineers to predict the performance of a design under various conditions, identify potential problems, and evaluate different design alternatives before physical prototypes are built.

Data Analysis: Statistical methods and probability theory are employed for analysing experimental data, validating models, and making predictions about the reliability and performance of designs. Statistical tools help engineers assess the uncertainty and variability in their measurements and make informed decisions based on data.

<u>Physics:</u>

Understanding Physical Principles: Physics provides the fundamental principles and laws that govern the behavior of natural phenomena. Engineers use principles from classical mechanics, thermodynamics, electromagnetism, and other branches of physics to understand how systems operate and interact with their environment.

Mechanical Design: In mechanical engineering, principles of statics, dynamics, and mechanics of materials are applied to design components and structures that can withstand loads, resist deformation, and perform their intended functions safely and efficiently.

Electrical Design: Electrical engineers rely on the principles of electromagnetism and circuit theory to design electronic devices, power systems, and communication networks. Understanding electrical properties such as voltage, current, resistance, and capacitance is essential for designing reliable and efficient electrical systems.

3. ***Explain the role of Prototyping in evaluating a Design.***

Prototyping plays a crucial role in evaluating a design by providing a tangible representation of the proposed solution that can be tested, refined, and validated before final implementation. Here are the key aspects of how prototyping contributes to the design evaluation process:

Visual Representation: Prototypes offer a visual representation of the design concept, allowing stakeholders to better understand how the final product will look and function. Visual prototypes can range from sketches and wireframes to computer-generated renderings or physical models, depending on the complexity of the design.

Functional Demonstration: Prototypes allow engineers and designers to demonstrate the functionality of the design in a realistic manner. Functional prototypes, which may include working mechanisms, electronics, or software simulations, enable stakeholders to interact with the design and provide feedback on its usability and performance.

Risk Reduction: Prototyping helps mitigate risks associated with the design by allowing engineers to validate key assumptions, test alternative solutions, and assess the feasibility of implementation. By identifying and addressing potential risks early on, such as technical challenges, manufacturing constraints, or market acceptance issues, prototyping helps minimize costly redesigns or failures later in the development process.

User Feedback Incorporation: Prototyping facilitates the gathering of user feedback and preferences through usability testing, focus groups, or surveys. By involving end-users in the evaluation process, designers can gain insights into user needs, preferences, and pain points, which can inform further refinements to the design to better meet user requirements.

Cost and Time Savings: While prototyping incurs upfront costs and time investment, it can ultimately save time and money by reducing the likelihood of costly errors or design changes during later stages of development or production. Identifying and resolving design issues early through prototyping can prevent expensive rework and delays down the line.

4. ***What are the factors considered in communicating designs through oral presentation?***

Audience Analysis: Understand your audience's level of technical expertise, interests, and preferences. Tailor your presentation style, language, and level of detail accordingly to ensure that the content resonates with your audience and effectively communicates the design.
Clarity and Conciseness: Clearly articulate the purpose, objectives, and key points of the design. Use concise language and avoid jargon or technical terms that may be unfamiliar to your audience. Structure your presentation in a logical manner with clear transitions between topics.
Visual Aids: Utilize visual aids such as slides, diagrams, charts, and prototypes to enhance understanding and engagement. Ensure that visual elements are clear, well-designed, and relevant to the content being presented.
Engagement Techniques: Engage your audience by maintaining eye contact, using gestures and body language, and varying your vocal tone and pace. Encourage interaction through questions, polls, or discussions to involve the audience and keep them attentive and engaged throughout the presentation.
Storytelling: Use storytelling techniques to illustrate the journey of the design process, highlighting key milestones, insights, and successes along the way.
Demonstration and Examples: If possible, provide live demonstrations or examples to showcase the functionality and features of the design. Use case studies, testimonials, or real-world scenarios to illustrate how the design addresses specific needs or solves problems for users.
Addressing Questions and Feedback: Anticipate questions that may arise during the presentation and be prepared to address them effectively. Encourage questions and feedback from the audience, and respond thoughtfully and respectfully to inquiries or comments. Use questions as opportunities to clarify points, deepen understanding, and foster discussion.
Time Management: Manage your time effectively to ensure that you cover all key points within the allotted time frame. Practice your presentation beforehand to gauge the timing and make adjustments as needed to fit within the time constraints.
Confidence and Professionalism: Project confidence, enthusiasm, and professionalism throughout your presentation. Speak clearly and confidently, maintain a positive demeanor, and demonstrate your expertise and passion for the design. Be prepared to handle unexpected challenges or technical issues calmly and professionally.

5. ***Describe how prototyping helps in design process***
   Prototyping helps in the design process by allowing designers to visualize and test their ideas in a tangible form before investing significant time and resources into full-scale production. By creating prototypes, designers can quickly identify potential flaws, refine their concepts, and iterate on their designs based on user feedback. Prototyping also facilitates communication and collaboration within design teams and with stakeholders, as it provides a common platform for discussing and evaluating design ideas. Ultimately, prototyping accelerates the design process by enabling designers to rapidly explore and validate multiple solutions, leading to more innovative and user-centric final designs.

6. ***What is visual communication in design***
   Visual communication in design refers to the use of images, graphics, and other visual elements to convey information, ideas, and messages. It encompasses various design

disciplines, including graphic design, web design, user interface design, and multimedia design. Visual communication is essential for effectively communicating complex concepts, enhancing user experience, and creating visually appealing designs that captivate and engage audiences. It involves the strategic use of color, typography, imagery, layout, and other visual elements to convey meaning and evoke emotions. In essence, visual communication in design aims to communicate information efficiently and creatively through visual means, enriching the overall communication experience for the viewer.

7. ***Explain any three advantages of communicating designs in writing.***

- **Clarity and Precision**: Describing designs in writing allows for clear and precise communication of ideas, specifications, and requirements. Through written descriptions, designers can articulate their concepts in detail, ensuring that all stakeholders have a comprehensive understanding of the design intent.
- **Documentation and Record-Keeping**: Written communication provides a permanent record of design decisions, revisions, and feedback throughout the design process. This documentation serves as a valuable reference for future iterations, enabling designers to track changes, address issues, and maintain consistency in design implementation.
- **Accessibility and Distribution**: Written design documentation can be easily shared and distributed among team members, clients, and other stakeholders, regardless of geographical location or time constraints. This accessibility facilitates collaboration, feedback, and decision-making, ensuring that all relevant parties are informed and involved in the design process.

8. ***How can a design be communicated through engineering sketches and drawings?***

Engineering sketches and drawings serve as visual communication tools to convey design concepts, technical specifications, and details effectively. Through sketches, designers can visualize and explore design ideas quickly, facilitating problem-solving and iteration in the design process. Detailed engineering drawings provide precise information about dimensions, tolerances, and materials, essential for manufacturing and construction. By sharing sketches and drawings, designers can communicate design intent, solicit feedback, and collaborate with team members and stakeholders, ensuring alignment and understanding throughout the design process. Overall, engineering sketches and drawings play a vital role in communicating designs, facilitating collaboration, and ensuring the successful realization of design projects.

9. ***"Sketching is a powerful tool in design engineering". Justify this statement.***

Sketching is a powerful tool in design engineering because it allows designers to visually explore and express their ideas quickly and intuitively. It facilitates communication and collaboration by providing a tangible representation of design concepts that can be easily shared and understood by team members and stakeholders. Sketching supports problem-solving and iteration by enabling designers to rapidly generate and evaluate multiple design alternatives, leading to more innovative and effective solutions. Additionally, sketching offers speed and efficiency in the design process, requiring minimal resources and technical skills while enabling designers to iterate rapidly and streamline the development of design concepts. Overall, sketching enhances creativity, communication, problem-solving, and efficiency in design engineering, making it an indispensable tool for designers.

10. ***What is a mathematical model? State the significance?***
    A mathematical model is a simplified representation of a real-world system or phenomenon using mathematical equations, formulas, or algorithms. It helps to describe, analyze, and predict the behavior or outcomes of the system under different conditions. The significance of mathematical models lies in their ability to provide insights into complex systems, make predictions, and guide decision-making processes. By abstracting real-world complexities into mathematical terms, models enable researchers, engineers, and policymakers to better understand the underlying mechanisms of systems, optimize processes, solve problems, and explore potential scenarios in a controlled and systematic manner. Mathematical models are widely used across various fields, including science, engineering, economics, biology, and social sciences, to improve our understanding of the world and inform evidence-based decision-making.

11. ***Write down the name of the product where (i) aesthetic is essential factor (ii) reliability is most important (iii) aesthetic is not considered***
    (i) Aesthetic is Essential Factor: Designer furniture or luxury watches.
    (ii) Reliability is Most Important: Aircraft engines or medical devices.
    (iii) Aesthetic is Not Considered: Industrial machinery or plumbing fixtures.

PART B Questions

12. ***Compare prototype of a car with its model***
13. ***Explain the method of developing a mathematical model for a passenger lift.***
14. ***Assume that you have completed the design of a new model ceiling fan and a prototype is needed for testing. Communicate your idea to the production department effectively to manufacture the prototype.***
15. ***Draw the detailed drawing of a foldable bed with design detailing, scale drawings and dimensions. Use hand sketches.***
16. ***Explain the steps for oral presentation for marketing your own product within limited budget? Illustrate with an example and suitable figures.***
17. ***Why do designers use mathematical modeling in the design process? Show an example of how mathematics and physics play a role in design.***
18. ***What are the different types of sketches used by designers to communicate their design? How will you graphically communicate the design of a creative coloring book with design detailing, material selection, scale drawings and dimension. Use hand sketches.***
19. ***Graphically communicate the design of a foldable ladder for electricians. Draw the detailed 2D drawings of the same with design detailing, material selection, scale drawings and dimensions. Use only hand sketches.***
20. ***Describe the role of mathematical modelling in design engineering. Show how mathematics and physics play a role in designing a simple paper cutting scissor.***
21. ***Explain the general guidelines for technical communication. Justify each point with an example.***
22. ***A web page has to be maintained to store the details of covid patients in Kerala district wise. Design a web page and its popups with neat sketches and necessary documentation. The design must include the specification of software required for the page development.***

23. ***Design an integrated water bottle with lunch box. Draw the detailed 2D drawings of the same with design detailing, material selection and dimensions. Use only hand sketches.***
24. ***Prepare a technical report for a newly designed portable ladder with neat sketches for presenting to a client.***
25. ***Design an office chair and communicate your design using sketches with design detailing, material selection, scale drawings and dimensions***
26. ***Describe the role of mathematical modelling in design engineering citing an example***
27. ***Design a foldable steel table. Draw the detailed 2D drawings of the same with design detailing, scale drawings and dimensions. Use only hand sketches.***
28. ***Prepare a technical report for a newly designed website for online training of students with neat diagrams for presenting to a client.***

Module – 4

1. ***Distinguish between project-based learning and problem-based learning in design engineering.***

Project-based Learning	Problem-based Learning
Project-based Learning begins with the assignment of tasks that will lead to the creation of a final product or artefact. The emphasis is on the end product. • Students work on open-ended assignments. These could be more than one problem. • Students analyse the problems and generate solutions. • Students design and develop a prototype of the solution. • Students refine the solution based on feedback from experts, instructors, and/or peers.	Problem-based Learning begins with a problem that determines what students study. The problem derives from an observable phenomena or event. The emphasis is on acquiring new knowledge and the solution is less important. • Students are presented with an open-ended, authentic question. • Students analyse the question • Students generate hypotheses that explain the phenomena. • Students identify further follow-up questions • Students seek additional data to answer the questions.

2. ***Discuss the role of life cycle design approach in design decisions.***

Environmental Impact Reduction: By identifying and assessing the environmental impacts associated with each stage of the product life cycle, designers can prioritize design strategies and technologies that minimize resource consumption, energy use, emissions, and waste generation. Life cycle design promotes the use of renewable materials, energy-efficient technologies, and pollution prevention measures to reduce the environmental footprint of products and systems.

Resource Efficiency: Life cycle design emphasizes resource efficiency by optimizing material and energy flows throughout the product life cycle. Designers seek to minimize the use of virgin materials, maximize the use of recycled or renewable materials, and reduce energy consumption and waste generation during manufacturing, transportation, and use. Design decisions that promote resource efficiency can lead to cost savings, improved competitiveness, and reduced environmental impact.

Product Durability and Longevity: Life cycle design encourages the development of products that are durable, long-lasting, and easy to maintain, repair, and upgrade. By extending the useful life of products and reducing the need for premature replacement or disposal, designers can minimize waste generation and environmental impact over time. Designing for durability and longevity can also enhance customer satisfaction, product reliability, and brand reputation.

End-of-Life Considerations: Life cycle design addresses end-of-life considerations by evaluating options for product reuse, recycling, remanufacturing, or disposal. Designers seek to design products that are recyclable, biodegradable, or easily disassembled for recycling or reuse of components and materials

Stakeholder Engagement and Collaboration: Life cycle design encourages collaboration and engagement with stakeholders across the supply chain, including suppliers, manufacturers, customers, and regulators. By involving stakeholders in the design process and considering their perspectives, concerns, and preferences, designers can develop more sustainable and socially responsible products that meet the needs and expectations of all stakeholders.

3. ***Illustrate advantages of reverse engineering in design.***

Understanding Legacy Systems: Reverse engineering allows engineers to understand and document legacy systems or products for which the original design documentation may be incomplete, outdated, or unavailable. By reverse engineering existing products, engineers can gain insights into their design principles, materials, manufacturing processes, and performance characteristics.

Product Improvement and Innovation: Reverse engineering enables engineers to identify opportunities for product improvement, innovation, and optimization. By analyzing existing products or components, engineers can identify design flaws, inefficiencies, or performance limitations and develop innovative solutions to enhance functionality, durability, or usability.

Competitive Analysis: Reverse engineering provides valuable insights into competitors' products, enabling companies to benchmark their own products against competitors and identify areas for differentiation and competitive advantage. By analyzing competitor products, companies can understand their strengths and weaknesses and develop strategies to enhance their market position.

Cost Reduction: Reverse engineering can help companies reduce costs by identifying alternative materials, manufacturing processes, or design approaches that achieve the same functionality at a lower cost. By reverse engineering expensive or proprietary components, companies can explore opportunities for cost-saving alternatives or negotiate better terms with suppliers.

Quality Assurance and Validation: Reverse engineering can be used to validate the quality and performance of products by comparing them to their original specifications or design intent. By reverse engineering products and conducting performance testing, companies can ensure that products meet quality standards, regulatory requirements, and customer expectations.

Customization and Adaptation: Reverse engineering allows companies to customize existing products or components to meet specific customer requirements or market demands. By reverse engineering off-the-shelf products, companies can adapt them to unique applications, environments, or user preferences, thereby expanding their market reach and customer base.

4. **Explain bio mimicry in design with an example**

Biomimicry, also known as biomimetics, is an approach to design that draws inspiration from nature to solve human challenges and create innovative solutions. It involves studying biological systems, processes, and structures and applying the principles and strategies observed in nature to design products, technologies, and systems.

Example: **Bullet Trains**

Principle: The bullet train is a high-speed train system developed in Japan. Engineers faced a challenge with the loud noise and air pressure changes that occurred when the train emerged from tunnels at high speeds. To address this issue, engineers turned to nature for inspiration.

Inspiration: **The kingfisher bird** dives into water with minimal disturbance, thanks to its long, narrow beak with a smooth, streamlined shape. Engineers observed that this shape allowed the kingfisher to move smoothly from air to water with minimal disruption.

5. ***Enumerate six features of a modular design***

- **Standardized Components**: Modular designs utilize standardized components that can be easily interchanged or replaced without requiring significant modifications to the overall system.
- **Scalability**: Modular designs allow for scalability, enabling systems to be easily expanded or reduced in size by adding or removing modules as needed.
- **Interchangeability**: Modules in a modular design are interchangeable, meaning that they can be used in different configurations or applications without modification.
- **Ease of Assembly and Disassembly**: Modular designs are characterized by ease of assembly and disassembly, allowing for rapid construction, maintenance, and repair of systems.
- **Flexibility**: Modular designs offer flexibility in design and configuration, allowing users to customize systems to meet specific requirements or preferences.
- **Ease of Maintenance and Upgrade**: Modular designs simplify maintenance and upgrade processes by allowing individual modules to be easily accessed, replaced, or upgraded without disrupting the entire system.

6. ***Concurrent engineering is better than sequential engineering- Justify***

Concurrent engineering, where different stages of product development occur simultaneously, is often considered superior to sequential engineering, where each stage is completed before moving on to the next. This is because concurrent engineering enables faster time-to-market by overlapping design, manufacturing, and testing phases, reducing overall development time. It promotes collaboration and communication between cross-functional teams, leading to better integration of design requirements and manufacturing constraints early in the process. Concurrent engineering also facilitates rapid iteration and problem-solving, as feedback from later stages can inform and improve earlier stages of development. Ultimately, by allowing for more efficient use of resources and fostering a more agile approach to product development, concurrent engineering offers greater flexibility and adaptability in responding to market demands and customer feedback.

7. ***Examine how engineering students can learn design engineering through projects.***

Engineering students can effectively learn design engineering through projects by engaging in hands-on, practical experiences that apply theoretical concepts to real-world problems. By working on projects, students gain valuable skills in problem-solving, critical thinking, and project management, as they navigate the entire design process from concept development to implementation. Projects provide opportunities for students to collaborate with peers, interact with industry professionals, and apply interdisciplinary knowledge to address complex engineering challenges. Additionally, feedback and reflection on project outcomes enable students to identify areas for improvement and refine their design skills, fostering continuous learning and professional development. Overall, project-based learning offers a dynamic and immersive learning experience that prepares engineering students for success in their future careers.

8. ***Distinguish how aesthetics and ergonomics change engineering designs.***

Aesthetics: Aesthetics in engineering design concern the visual appearance and appeal of a product. Aesthetically pleasing designs consider factors such as color, shape, texture, and overall styling to create products that are visually attractive and appealing to users. Aesthetics can influence consumer perceptions, brand identity, and user satisfaction, as well as differentiate products in the marketplace.

Ergonomics: Ergonomics, on the other hand, focuses on designing products and systems to optimize human performance and well-being. Ergonomic design considers factors such as comfort, usability, safety, and efficiency to ensure that products are comfortable and easy to use for intended users. Ergonomics aims to minimize physical strain, reduce the risk of injury, and enhance user experience by aligning product design with human capabilities and limitations.

In summary, while aesthetics primarily address the visual aspects of design to enhance the appeal of products, ergonomics focus on optimizing usability and user experience to improve functionality, comfort, and safety. Both aesthetics and ergonomics contribute to the overall quality and effectiveness of engineering designs, but they address different aspects of the design process and user interaction.

9. ***Explain life cycle design.***

Life cycle design, also known as life cycle assessment (LCA) or cradle-to-grave analysis, is an approach used in engineering and product design to evaluate the environmental impact of a product or system throughout its entire life cycle. This includes all stages from raw material extraction and manufacturing to use, maintenance, and disposal or recycling. Life cycle design aims to minimize environmental impacts by considering factors such as energy consumption, resource depletion, emissions, waste generation, and ecological footprint at each stage of the product life cycle. By analyzing the environmental consequences of design decisions and identifying opportunities for improvement, life cycle design helps engineers and designers make more sustainable choices and develop products that have minimal environmental impact over their entire life cycle.

10. ***Describe how aesthetics is important in design process***

- **User Appeal**: Aesthetics influence how users perceive and interact with products. A visually appealing design can attract users, evoke positive emotions, and enhance their overall experience.
- **Brand Image**: Aesthetics contribute to brand identity and differentiation. Well-designed products communicate quality, professionalism, and attention to detail, strengthening brand perception and customer loyalty.
- **Market Competitiveness**: In competitive markets, aesthetics can be a key differentiator. Products with attractive designs stand out from competitors and may command higher prices or market share.
- **Functional Integration**: Aesthetics are not just about looks; they also influence usability and ergonomics. A well-designed product considers both form and function, balancing aesthetic appeal with practicality and user needs.
- **Innovation and Creativity**: Aesthetics encourage creativity and innovation in design. Designers often push boundaries and explore new ideas to create visually striking and unique products that capture attention and inspire others.

11. ***Compare sequential design and modular design techniques?***

Sequential Design:	**Modular Design**:

• **Sequential Process**: In sequential design, the product development process follows a linear sequence of stages, where each stage is completed before moving on to the next. For example, design, manufacturing, testing, and marketing are typically carried out sequentially. • **Rigidity**: Sequential design tends to be more rigid and inflexible, as changes or iterations in one stage may require revisiting earlier stages, leading to delays and increased costs. • **Specialization**: Sequential design allows for specialization, as each stage can be assigned to specialized teams or departments with specific expertise in that area. • **Clarity**: Sequential design offers clarity and structure, as the progression from one stage to the next is well-defined and easy to follow.	• **Parallel Development**: In modular design, different modules or components of the product are developed concurrently, allowing for parallel development and faster time-to-market. • **Flexibility**: Modular design offers greater flexibility and adaptability, as changes or updates can be implemented more easily by replacing or modifying individual modules without affecting the entire system. • **Integration**: Modular design promotes integration and collaboration across teams, as different modules need to interface and work together seamlessly. • **Scalability**: Modular design enables scalability, as additional modules can be added or removed to customize the product for different applications or requirements.

12. ***Describe the use of value engineering in the design process.***

    Value engineering, also known as value analysis or value management, is a systematic approach used in the design process to optimize the value of a product or system while minimizing costs. It involves analyzing the functions and performance requirements of a design and identifying opportunities to improve value by enhancing functionality, reducing costs, or increasing efficiency. Value engineering focuses on achieving the desired outcomes at the lowest possible cost without sacrificing quality or performance. By emphasizing value and cost-effectiveness, value engineering helps designers make informed decisions, prioritize design choices, and maximize the return on investment throughout the design process.

PART B Questions

13. ***Apply value engineering to a pen, and design a lightweight pen torch. Illustrate the solution using sketches.***
14. ***Design waste bins to be kept at bus stops for waste collection enabling source separation. The bin should be theft-resistant and protect the contents of the bin from external weather conditions. Design the bins with ergonomic consideration for waste collection workers. Sketch the design using hand drawings.***
15. ***Illustrate modular design approach for designing of desktop computers***
16. ***Demonstrate the concept of ergonomics through design of a table lamp***
17. ***Show the development of a nature-inspired design for a fashionable umbrella based on a banana leaf. Use hand sketches to support your arguments.***
18. ***Develop some design modification for sports utility bag, to improve its functionalities as well as product value. Sketch the design.***

19. ***Design a nature inspired solar lamp for the students residing in urban areas. These students do not have proper availability of electricity and cannot afford highly priced products. Illustrate your design with sketches.***
20. ***Apply modular engineering to a conventional bicycle and design a bicycle which can used in different terrains. Illustrate the design using sketches.***
21. ***What is meant by modular design? Apply the modular design concept to design a bicycle.***
22. ***What is the importance of project-based learning and problem-based learning? Use project based learning method to design a modern city for the year 2030.***
23. ***Show the design of a kid friendly craft and then depict how the design changes when considering 1) aesthetics and 2) ergonomics into consideration. Give hand sketches and explanations to justify the changes in designs.***
24. ***Design a smart umbrella which has automatically folding and unfolding mechanism during heavy rain and sunlight with the concept of bio mimicry and sustainability. Use hand sketches to support your arguments.***
25. ***Illustrate the modular design of a dining table where three different geometrical varieties can be developed as per the customer requirement.***
26. ***Explain the aesthetic, ergonomic and safety considerations incorporated in the design of a baby tricycle.***

Module – 5

1. ***Show how designs are varied based on the aspects of production methods, life span, reliability, and environment?***

Production Methods:

Traditional Manufacturing: Designs for products manufactured using traditional methods, such as injection molding or casting, may prioritize simplicity and ease of manufacturing. Complex shapes or intricate details may be challenging or costly to produce.

Additive Manufacturing (3D Printing): Designs for products produced using additive manufacturing methods, like 3D printing, may leverage the freedom of design afforded by layer-by-layer construction. This can enable the creation of complex geometries, lightweight structures, and customized components.

CNC Machining: Designs for products fabricated through CNC machining may focus on optimizing tool paths and material usage to minimize waste and machining time. Features requiring high precision may be emphasized, and material selection may consider machinability.

Hybrid Manufacturing: Designs for products made using hybrid manufacturing processes, which combine additive and subtractive methods, may integrate features optimized for both processes. For example, additive methods may be used for complex geometries, while subtractive methods may be employed for finishing surfaces.

Life Span:

Short Life Span: Designs for products with a short life span, such as consumer electronics or fashion items, may prioritize cost-effectiveness, aesthetics, and current trends over long-term durability or repairability.

Long Life Span: Designs for products intended to have a long life span, such as industrial machinery or infrastructure, may prioritize durability, reliability, and ease of maintenance. Materials and components may be selected for their longevity and resistance to wear and degradation over time.

Reliability:

High Reliability: Designs for mission-critical or safety-critical applications, such as medical devices or aerospace components, may prioritize reliability and fault tolerance. Redundant systems, rigorous testing, and robust materials may be incorporated to minimize the risk of failure.

Moderate Reliability: Designs for products where reliability is important but not critical, such as household appliances or automotive components, may balance reliability with cost and performance considerations. Design features may be implemented to mitigate common failure modes and extend service life.

Low Reliability: Designs for products where reliability is less critical, such as disposable consumer goods or promotional items, may prioritize cost and aesthetics over reliability. Simplified designs and lower-cost materials may be utilized to meet price targets.

Environment:

Harsh Environments: Designs for products intended for use in harsh environments, such as marine or outdoor applications, may prioritize resistance to corrosion, UV degradation, temperature extremes, and moisture ingress. Materials and coatings may be selected for their ability to withstand environmental stressors.

Controlled Environments: Designs for products used in controlled environments, such as laboratory equipment or indoor electronics, may focus on performance, precision, and cleanliness. Materials may be chosen for their compatibility with cleanroom standards or specific operating conditions.

Sustainable Design: Designs for products aimed at reducing environmental impact may prioritize sustainability, energy efficiency, and recyclability. Design considerations may include material selection, eco-friendly manufacturing processes, product life cycle analysis, and end-of-life disposal or recycling options.

2. ***Explain how economics influence the engineering designs?***

Cost Considerations: Economics heavily influence design decisions, as engineers must balance performance and functionality requirements with cost constraints. Designing products that are cost-effective to manufacture, assemble, and maintain is essential for achieving profitability and competitiveness in the market.

Value Engineering: Value engineering is a systematic approach that aims to optimize the value of a product or system by maximizing performance while minimizing costs. Engineers analyze the function of each component and explore alternative design options to achieve the desired functionality at the lowest possible cost.

Mass Production vs. Customization: Economics influence the choice between mass production and customization in engineering design. Mass-produced products typically benefit from economies of scale, allowing for lower per-unit costs. However, customization may be preferred in certain markets or applications where personalized solutions command higher prices or offer competitive advantages.

Supply Chain Optimization: Engineering designs are influenced by considerations related to the supply chain, including sourcing of materials, manufacturing processes, transportation costs, and inventory management. Design decisions may be influenced by factors such as availability of raw materials, labor costs, shipping distances, and lead times.

Life Cycle Cost Analysis: Engineers often conduct life cycle cost analysis to evaluate the total cost of ownership of a product over its entire life cycle, including initial acquisition costs, operating costs, maintenance costs, and disposal costs. Design decisions may be guided by considerations of long-term cost-effectiveness and return on investment.

Market Demand and Pricing: Economics influence engineering designs through considerations of market demand, pricing strategies, and competitive positioning. Engineers may design products that align with market preferences, meet customer needs, and offer features and performance levels that justify their price point relative to competitors.

Resource Efficiency and Sustainability: Economics influence engineering designs by driving considerations of resource efficiency, sustainability, and environmental impact. Design decisions may prioritize the use of renewable materials, energy-efficient technologies, and eco-friendly manufacturing processes to reduce costs, minimize waste, and mitigate environmental risks.

3. ***Describe ethics to be followed in engineering design***

Safety: Engineers have a moral obligation to prioritize the safety and welfare of the public, users, and stakeholders affected by their designs. They should ensure that products, systems, and structures are designed to minimize risks of injury, harm, or damage during operation or use.

Health and Well-being: Engineers should consider the potential impact of their designs on the health and well-being of individuals and communities. They should strive to create products and systems that promote human health, accessibility, inclusivity, and quality of life for all users.

Environmental Sustainability: Engineers should incorporate principles of environmental sustainability into their design practices to minimize negative impacts on the environment and natural resources. They should strive to reduce pollution, conserve energy, minimize waste, and promote ecological stewardship throughout the product life cycle.

Social Responsibility: Engineers should consider the broader social, cultural, and ethical implications of their designs on society. They should respect diversity, equity, and human rights, and strive to create inclusive and socially responsible solutions that benefit communities and promote social justice.

Transparency and Accountability: Engineers should be transparent and accountable for their design decisions and actions. They should honestly communicate the capabilities, limitations, and potential risks associated with their designs, and be willing to address concerns or criticisms raised by stakeholders.

Compliance with Laws and Regulations: Engineers should comply with applicable laws, regulations, codes, and standards governing their profession and the industries in which they operate. They should ensure that their designs meet legal requirements, safety standards, environmental regulations, and ethical guidelines.

4. ***Explain the significance of sustainability in engineering design***

Environmental Preservation: Engineering activities have significant environmental implications, including resource depletion, pollution, habitat destruction, and climate change. Sustainable engineering design aims to minimize these negative impacts by reducing resource consumption, emissions, and waste generation, thereby preserving natural ecosystems and biodiversity.

Resource Conservation: Sustainable engineering design emphasizes the efficient use of resources, including energy, water, materials, and land. By optimizing resource utilization and promoting recycling, reuse, and renewable alternatives, engineers can minimize resource depletion and promote long-term resource availability for future generations.

Mitigation of Climate Change: Engineering design plays a critical role in mitigating climate change by reducing greenhouse gas emissions and promoting renewable energy sources. Sustainable design practices such as energy efficiency improvements, carbon footprint reduction, and low-carbon technologies help combat climate change and contribute to a more sustainable future.

Social Equity and Justice: Sustainable engineering design incorporates principles of social equity and justice by addressing the needs of marginalized communities, promoting access to essential services, and reducing disparities in resource distribution. Designs that prioritize inclusivity, affordability, and accessibility contribute to a more equitable and just society.

Economic Viability: Sustainable engineering design can offer economic benefits by reducing operating costs, improving efficiency, and enhancing long-term resilience. Investments in sustainability can lead to cost savings, innovation opportunities, and competitive advantages for businesses and organizations, thereby promoting economic growth and prosperity.

Regulatory Compliance and Risk Management: Sustainable engineering design helps organizations comply with environmental regulations, standards, and best practices, reducing legal and reputational risks associated with non-compliance. By proactively addressing environmental and social risks, engineers can mitigate liabilities and safeguard the interests of stakeholders.

5. ***How to estimate the cost of a particular design?***

- **Material and Component Costs**: Break down the design into its constituent materials and components, and research the current market prices for each. Consider factors such as quantity, quality, and sourcing options to determine the cost of materials.
- **Labor and Manufacturing Costs**: Estimate the labor hours required to manufacture each component of the design, and multiply by the appropriate labor rates to calculate labor costs. Consider overhead costs, such as facility expenses and equipment depreciation, to determine manufacturing costs.
- **Tooling and Equipment Costs**: Factor in any tooling or equipment required for manufacturing, such as molds, dies, or machinery. Estimate the cost of acquiring, maintaining, or leasing this equipment over the design's lifecycle.
- **Prototype and Testing Costs**: Include costs associated with prototyping, testing, and validation of the design to ensure functionality, safety, and compliance with standards. This may involve expenses for materials, labor, equipment, and testing facilities.
- **Contingency and Overhead**: Add a contingency allowance to account for unforeseen expenses or fluctuations in material and labor costs. Additionally, include overhead costs such as administrative expenses, marketing, and profit margin to determine the total cost of the design.

6. ***How do ethics play a decisive role in engineering design?***

Ethics play a decisive role in engineering design by guiding engineers to make morally sound decisions and prioritize the well-being of society, the environment, and future generations. Engineers have a responsibility to uphold ethical principles such as integrity, honesty, transparency, and accountability throughout the design process. Ethical considerations influence design choices, ensuring that products and systems are safe, reliable, and sustainable, and that they comply with legal and regulatory requirements. By integrating ethical considerations into engineering design, engineers can mitigate risks, prevent harm, and contribute to the greater good, fostering trust, credibility, and social responsibility within the engineering profession.

7. ***What are design rights, and how can an engineer put it into practice?***

Design rights refer to legal protections granted to the appearance, shape, or configuration of a product or article resulting from the design process. These rights typically include design patents, registered designs, or design copyrights, which provide exclusive rights to the creator to use, reproduce, or license the design for a specified period. Engineers can put design rights into practice by proactively seeking protection for their original designs through appropriate legal mechanisms. This involves documenting and formally registering the design with relevant intellectual property offices to establish ownership and prevent unauthorized copying or infringement by competitors. By securing design rights, engineers can safeguard their intellectual property, maintain a competitive edge, and maximize the commercial value of their designs in the marketplace.

8. ***Explain the cost factor calculation of a particular design?***

- **Material Costs**: Estimate the quantity and cost of materials required for the design. This includes raw materials, components, and any purchased parts or supplies.

- **Labor Costs**: Determine the labor hours needed to manufacture, assemble, and test the design. Multiply the labor hours by the appropriate labor rates to calculate labor costs.

- **Overhead Costs**: Consider overhead expenses such as facility rent, utilities, equipment depreciation, and administrative costs. Allocate a portion of these costs to the design based on factors like space usage and labor hours.

9. ***What is sustainable engineering***

Sustainable engineering is an approach to engineering that seeks to design, develop, and implement solutions that meet the needs of the present without compromising the ability of future generations to meet their own needs. It involves integrating principles of environmental responsibility, social equity, and economic viability into engineering practices and decision-making processes. Sustainable engineering aims to minimize negative environmental impacts, conserve natural resources, promote social equity and justice, and enhance economic prosperity. By adopting sustainable engineering principles, engineers strive to create innovative solutions that address global challenges such as climate change, resource depletion, pollution, and social inequality, while promoting long-term sustainability and resilience.

10. ***Show how designs are finalized based on the aspects of production methods.***

Designs are finalized based on production methods through a systematic evaluation of factors such as manufacturability, cost-effectiveness, and efficiency. Engineers assess the feasibility of producing the design using available production methods, considering factors such as material compatibility, tooling requirements, and production capacity. They optimize the design to streamline manufacturing processes, minimize waste, and reduce production costs while ensuring product quality and performance. By aligning the design with appropriate production methods, engineers ensure that the final product can be efficiently and economically manufactured at scale, meeting both technical requirements and business objectives.

11. ***Tell how ethics play a decisive role in engineering design.***

Ethics play a decisive role in engineering design by guiding engineers to make morally sound decisions and prioritize the well-being of society, the environment, and future generations. Engineers have a responsibility to uphold ethical principles such as integrity, honesty, transparency, and accountability throughout the design process. Ethical considerations influence design choices, ensuring that products and systems are safe, reliable, and sustainable, and that they comply with legal and regulatory requirements. By integrating ethical considerations into engineering design, engineers can mitigate risks, prevent harm, and contribute to the greater good, fostering trust, credibility, and social responsibility within the engineering profession.

12. ***List the factors affecting the cost of a product***
   - **Material Costs:** Estimate the quantity and cost of materials required for the design. This includes raw materials, components, and any purchased parts or supplies.
   - **Labor Costs**: Determine the labor hours needed to manufacture, assemble, and test the design. Multiply the labor hours by the appropriate labor rates to calculate labor costs.
   - **Overhead Costs**: Consider overhead expenses such as facility rent, utilities, equipment depreciation, and administrative costs. Allocate a portion of these costs to the design based on factors like space usage and labor hours.

13. ***Enumerate the features of a sustainable product***
   - **Environmentally Friendly Materials**: Sustainable products are made from renewable, recyclable, or biodegradable materials that minimize environmental impact throughout their lifecycle.
   - **Energy Efficiency**: Sustainable products are designed to consume less energy during manufacturing, use, and disposal phases, reducing greenhouse gas emissions and resource depletion.
   - **Minimal Waste**: Sustainable products are designed to minimize waste generation during production and use, promoting efficient use of resources and reducing pollution.
   - **Repairability and Upgradability**: Sustainable products are designed for easy repair, maintenance, and upgrades, allowing users to extend product lifespan and reduce the overall environmental footprint.
   - **Social Responsibility**: Sustainable products are produced under fair labor practices, ensuring safe working conditions, fair wages, and respect for human rights throughout the supply chain.
   - **Minimal Environmental Impact**: Sustainable products aim to minimize environmental impact across all stages of the product lifecycle, including raw material extraction, manufacturing, transportation, use, and disposal.

PART B Questions

14. ***Describe how to estimate the cost of a chair. Identify the influence of cost on engineering design. Explain the difference in the design of low cost chair and a high quality chair.***
15. ***Explain the impact of electronic gadgets on environment and how to minimise the impact with design modification***
16. ***Examine the changes in the design of a hand purse with constraints of 1) production methods 2) life span requirement 3) reliability issues and 4) environmental factors. Use hand sketches and give proper rationalization for the change in design***
17. ***Describe how to estimate the cost of a particular design of a smart bus for public use with Wi-Fi and proper cleaning facility. List the various components used. Show how economics will influence the engineering designs, use hand sketches to support the arguments.***
18. ***Describe the importance of evaluating economic considerations in product development with an example.***
19. ***Illustrate the changes in design of a solar powered street light in terms of production, use, and sustainability with the help of sketches***

20. *Examine the changes in the design of a footwear in terms of production methods, reliability issues and environmental factors. Use hand sketches and give proper rationalization for the changes in design.*
21. *Describe how to estimate the cost of a residence house in design stage. Show how the economics will influence its design. Use hand sketches to support your arguments.*
22. *A table has to be designed as a study table, but it must include a provision to place your computer. Calculate the cost difference if you want to convert it as a dining table. The cost calculation must include labor, material and over-head costs.*
23. *An umbrella has to be designed for daily use. Show how the cost will vary based on material, design and labor using suitable neat hand sketches?*
24. *Design a sustainable piping network for reuse of water in a residential building enabling water conservation. Sketch the design.*
25. *Design a door handle with a lock which is easy to use. Use hand sketches and give rationalization for the various features in the design.*
26. *Illustrate the changes in design of disposable tea cup in terms of production, use and sustainability*
27. *Describe the how to estimate the cost of a table in design stage? Show how economics will influence the engineering designs.*
28. *Design a fan which automatically reduces speed or stops when the temperature reduces during the night for energy conservation. Use hand sketches to support your design.*
29. *Describe how to estimate the cost of a pen and list the various parts. Show how the economics will influence the engineering designs. Use hand sketches to support your arguments.*